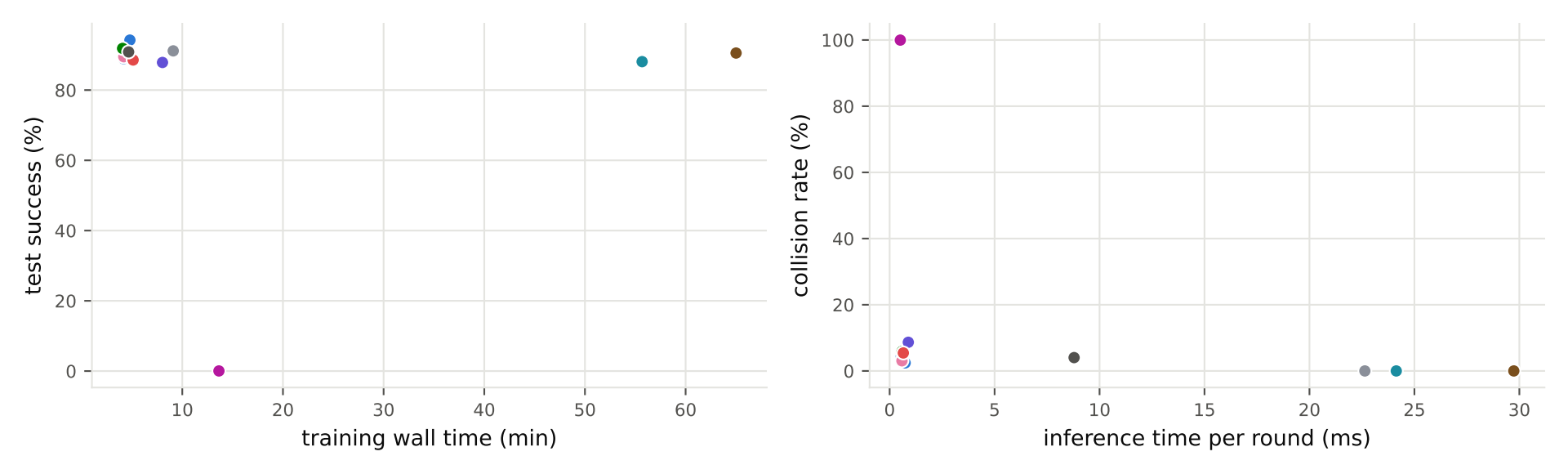
test success (%)
0
20
40
60
80
10
20
30
40
50
60
training wall time (min)
collision rate (%)
0
20
40
60
80
100
0
5
10
15
20
25
30
inference time per round (ms)
DDPG (paper)
+ TD3
+ HER
+ OBL
+ APF shaping
+ APF prior (residual)
+ CBF shield
+ set encoder (all pairs)
+ TDA curriculum (on MLP full stack)
CEM-RL (population) on TD3+HER